Introduction to Brain & Cognition

Comprehensive Lecture Notes

Lectures: March 4, 6, 10, 13, 17, 24, 27 — 2026

IIIT Hyderabad

Contents

Lecture 1: The Science of the Mind (March 4)

1.1 Course Overview

This course provides a broad knowledge and appreciation of **cognitive psychology**—the scientific study of how the mind acquires, stores, retrieves, and uses knowledge to influence behaviour. The course draws from two textbooks:

1. *Cognition: Exploring the Science of the Mind* (7th edition)
2. *Thinking 101* (based on a popular Yale course)

Cognition

Cognition (Latin: *cognoscere* — "to think, to know") refers to the mental processes by which knowledge is acquired, stored, retrieved, and used. It encompasses perception, attention, memory, language, reasoning, problem-solving, and decision-making.

1.2 Psychology and its Branches

Psychology is the study of the *psyche* (mind). Major branches include:

- **Experimental Psychology** — uses experimental methods to understand mind and behaviour
- **Social Psychology** — how groups behave together; group mind and group behaviour
- **Physiological Psychology** — what brain areas are important for certain behaviours; the material basis of behaviour (neurotransmitters, brain regions)
- **Cognitive Psychology** — how knowledge influences behaviour; how knowledge is acquired, stored, and retrieved

1.3 Historical Epochs of Psychology

1.3.1 Epoch 1: Introspection (1800s)

Introspection

Introspection is the method of studying the mind by looking inward—observing one's own thoughts, feelings, and mental processes, and trying to identify patterns. Pioneered by **William James** (two volumes based on introspection and observation).

Introspection borders on philosophy and psychology. It involves reflecting on one's own thought processes: "How do I make decisions? How do I perceive objects?"

Limitations of Introspection

1. **Inconsistency** — the same question can yield different answers on different occasions
2. **Individual variability** — different people introspect and arrive at different conclusions
3. **Bias** — constrained by one's own experience and knowledge
4. **Cannot access unconscious processes** — introspection only works when you are awake and conscious
5. **Non-verifiable** — "you are the observer, the witness, and the judge"
6. Violates scientific requirements of *repeatability* and *verifiability*

1.3.2 Epoch 2: Behaviourism (∼1900s–1950s)

Behaviourism

Behaviourism is a school of psychology that focuses exclusively on *observable behaviour*, dismissing the concept of a non-physical mind. All cognition is reduced to **stimulus–response (S–R) associations**. Prominent figure: **John B. Watson**.

Stimulus (Input) $\xrightarrow{\textit{Association}}$ **Response** (Output)

Key principles of behaviourism:

- Behaviour types: **Avoidance behaviour** (retract from stimulus) and **Approach behaviour** (move toward stimulus)
- Stimuli can be from any input modality: visual, auditory, tactile, olfactory
- Repeated stimulus–response pairing strengthens the association (forming **habits**)
- Cognition is split into: (a) *Wired/genetic* (innate reflexes) and (b) *Learnt* (acquired S–R associations)
- "There is nothing called mind"—only reflexes and associations

Built-in reflexes (genetic):

- Eye-blink reflex (puff of air → blink)
- Withdrawal reflex (hand touches hot surface → retract)
- Crying when hungry (in infants)

Limitations of Behaviourism

1. **Stimulus generalisation problem** — "Pass the salt" vs. "Salt, please!" vs. "Could you pass the salt?" yield the same response, yet the stimuli are different. Behaviourism cannot explain how internal meaning mediates equivalent responses.
2. **Novel stimuli** — humans respond correctly to requests they have *never* heard before. Where does this generalisation come from if not from an internal "mind"?
3. **Cannot explain language acquisition** — children acquire language with very few examples, too fast for pure S–R learning.
4. **Cannot explain unconscious processing** or internal representations.

1.3.3 Epoch 3: The Cognitive Revolution (∼1950s–60s)

Cognitive Revolution

The **Cognitive Revolution** was the paradigm shift from behaviourism to **cognitivism**, reintroducing the concept of the **mind** as an internal entity that mediates behaviour. The mind is no longer a taboo word.

Key contributors:

Edward Tolman — Cognitive Maps. Tolman studied rat navigation in mazes. Rats could generalise to *new paths* they had never visited, finding food efficiently. This implied an internal **cognitive map**—a stored representation useful in novel scenarios, not explainable by pure S–R.

George Miller — Chunking and Working Memory.

Miller's Law: 7 ± 2

Working memory (the temporary buffer) has a limited capacity of approximately 7 ± 2 **chunks**. We overcome this limit by **chunking**—compressing information into meaningful units.
Example: A 10-digit phone number exceeds capacity. We break it into chunks: area code (chunk 1) + first group (chunk 2) + second group (chunk 3) = 3 chunks ≤ 7.

Working Memory vs. Long-Term Memory

Working memory is a temporary scratch-pad with limited capacity (7 ± 2 chunks). Used for tasks like mental arithmetic.
Long-term memory can store vast amounts of information (e.g., entire poems) but requires repeated rehearsal and rote learning.

Noam Chomsky — Universal Grammar.

- Published a seminal book attacking behaviourism
- Argued that S–R associations *cannot* explain how children acquire language so rapidly from so few examples
- Proposed **Universal Grammar (UG)**: every brain is born with an innate, general-purpose grammar machine
- Exposure to a particular language (Telugu, Hindi, German, etc.) *specialises* this UG
- Also contributed the hierarchy of formal languages: context-free, context-sensitive grammars, etc.

1.4 Schemas and Scripts

Schema / Script

A **schema** (or **script**) is a mental framework—a compressed, structured representation of typical experiences—that helps us organise knowledge, generate expectations, and navigate the world in real time.

Example — Restaurant schema:

1. Enter restaurant → waiter asks how many people
2. Taken to a table → given a menu
3. Order food → waiter writes it down and reads it back
4. Food arrives → eat → pay bill

Example — Piggy bank schema:

"She shook it. It made no sound. She went to look for her mother."

To interpret this, we need the schema: children receive money → store in piggy bank → shaking reveals contents by sound → no sound = no money → go to mother for money. Without this schema, the sentences are incomprehensible.

Counterfactual Reasoning

Counterfactual reasoning: "If X were the case, what would the consequence be?" This higher-order reasoning is largely unique to humans and is not well explained by S–R models. Animals (except possibly great apes) show limited counterfactual reasoning.

1.5 Real-Time Constraint on Cognition

Operating in the real world imposes severe **time constraints** on cognitive processing:

- A cricket batsman has only ∼15–50 ms to decide how to play a delivery (full-toss, swing, etc.)
- Schemas and compressed representations allow rapid decision-making under time pressure

1.6 Research Methods in Cognitive Psychology

Cognitive psychology uses multiple methods:

1. **Behavioural experiments** — controlled lab tasks measuring accuracy, reaction time
2. **Eye tracking** — tracks position and movement of the eye; pupil dilation indicates cognitive load
3. **Physiological measurements** — galvanic skin conductance (GSR), heart rate variability (HRV)
4. **Neuroimaging** — EEG, fMRI, MEG, PET (detailed in later lectures)
5. **Computational modelling** — mathematical models of cognitive processes

Pupil Dilation and Cognitive Load

People engaged in *complex* tasks (e.g., 5-digit × 4-digit multiplication) show **pupil dilation**—an internal physiological reaction reflecting increased attention and cognitive effort. Simple tasks show no such dilation.

Practice & Review

Q1. [Multiple Choice] Which best defines *cognition*?

(A) The study of brain physiology (B) How knowledge is acquired, stored, retrieved, and used to guide behaviour
(C) Observable stimulus–response behaviour only (D) Signal processing in the peripheral nervous system

Answer: **B.** Cognition (Latin *cognoscere*: "to know") encompasses perception, attention, memory, language, and reasoning. It is the study of how knowledge is acquired, stored, retrieved, and used.

Q2. [Short Answer] What is introspection and what are four of its key limitations?

Answer: Introspection is the method of studying the mind by observing one's own thoughts, feelings, and mental processes. Limitations: (1) **Inconsistency**—the same question yields different answers on different occasions; (2) **Individual variability**—different people arrive at different conclusions; (3) **Bias**—constrained by personal experience and knowledge; (4) **No access to the unconscious**—introspection only operates when conscious; (5) **Non-verifiable**—the introspecting person is simultaneously observer and judge, violating scientific repeatability.

Q3. [Essay] Explain behaviourism, its core claim, and the three problems identified during the Cognitive Revolution. Reference Tolman, Miller, and Chomsky.

Answer: Behaviourism (Watson): all cognition reduces to stimulus–response (S–R) associations formed by repeated pairing; "mind" is inadmissible as a scientific concept. Key failures: (1) **Stimulus generalisation**—"Pass the salt," "Salt, please," and "Could you pass the salt?" all elicit the same response even though the stimuli differ; S–R theory cannot explain the internal *equivalence mapping.* (2) **Novel stimuli**—humans respond appropriately to requests never heard before. (3) **Language acquisition**—children acquire language too rapidly and from too few examples for pure S–R learning. Cognitive Revolution responses: **Tolman** demonstrated rats use internal *cognitive maps* to navigate new maze paths, implying stored internal representations; **Miller** established that working memory has a capacity of 7 ± 2 chunks (chunking = cognitive compression); **Chomsky** proposed Universal Grammar—an innate, general-purpose grammar machine specialised by early language exposure.

Q4. [Multiple Choice] Who proposed the theory of Universal Grammar?

(A) William James (B) Noam Chomsky (C) Edward Tolman (D) B.F. Skinner

Answer: **B.** Chomsky proposed that every brain is born with a universal grammar machine. Early exposure to a specific language (e.g., Telugu, Hindi, German) specialises it for that language's rules and sounds.

Q5. [Short Answer] State Miller's Law (7 ± 2) and explain how chunking allows a 10-digit phone number to be stored in working memory.

Answer: Working memory holds 7 ± 2 *chunks* of information. A raw 10-digit number (e.g., 9440055123) exceeds this limit. Chunking compresses it: area code **944** + group **0055** + group **123** = 3 chunks—well within the 7 ± 2 capacity. Chunking is a cognitive compression strategy that allows the limited buffer to store more information.

Q6. [Short Answer] What is a schema? Use the "piggy bank" example to explain why schemas are *necessary* for comprehending: *"She shook it. It made no sound. She went to look for her mother."*

Answer: A schema (or script) is a compressed mental framework of typical experiences used to organise knowledge and generate expectations. The piggy bank schema: children receive money as gifts → store coins in a piggy bank → shaking reveals contents by sound → silence means empty → seek money from a parent. Without this schema the three sentences are individually comprehensible but their causal connection—and the inference that the child needs money—is impossible to derive.

Q7. [Multiple Choice] Counterfactual reasoning involves:

(A) Recalling a specific past episode in detail
(B) Imagining a hypothetical state of affairs and reasoning about its consequences
(C) Forming a classical stimulus–response association
(D) Recognising a familiar face in a crowd

Answer: **B.** Counterfactual reasoning ("if X were the case, what would happen?") is a higher-order cognitive skill largely unique to humans; non-human animals show little evidence of it. It cannot be explained by S–R theory because there is no physical stimulus to respond to.

Q8. [Short Answer] Name four research methods used in cognitive psychology and state what each specifically measures.

Answer: (1) **Behavioural experiments**—accuracy and reaction time (RT) in controlled lab tasks; (2) **Eye tracking**—gaze position, fixation duration, and pupil dilation (a physiological proxy for cognitive load—pupil dilates for complex tasks); (3) **Physiological recording**—galvanic skin conductance (GSR) for arousal, heart rate variability (HRV) for autonomic state; (4) **Neuroimaging**—brain activity patterns via EEG, fMRI, MEG, or PET.

Lecture 2: Neural Basis for Cognition — Methods (March 6)

2.1 Recap: Three Epochs and the Fourth

Epoch	Approach	Era	Key Idea
1	Introspection	1700s–1800s	Look inward; subjective
2	Behaviourism	Early 1900s–1950s	S–R associations; objective
3	Cognitive Revolution	1950s–60s	Mind as mediator; internal representations
4	Consciousness	Emerging	Subjective experience; qualia

2.2 Attention

Attention

Attention is the cognitive ability to selectively focus on specific stimuli while inhibiting others. It enables prioritisation of relevant information.

The Cocktail Party Problem

In a noisy party with many simultaneous conversations, you can **focus on one speaker** and inhibit all others. This is **focused/selective attention**—a fundamental cognitive function.

Cognitive functions studied: perception, attention, inference, judgment, emotional feelings (happy, sad, angry), language, pattern recognition.

2.3 Consciousness and Qualia

Consciousness

Consciousness is the state of being aware of one's surroundings, thoughts, and emotions. It involves:

1. Being alive, awake, and able to respond to stimuli
2. The ability to **experience and reflect** on those experiences
3. Using reflection to guide future actions

Self-consciousness (the ability to reflect on one's own experience) is argued to be uniquely human.

Qualia

Qualia (singular: *quale*) refer to the *subjective, personal* quality of conscious experiences. E.g., the "crunchiness" and "sweetness" you experience when biting an apple is *your* personal quale—it cannot be directly shared or observed by others.

Consciousness as the 4th Epoch

Current science does not have a complete framework for consciousness. Like "mind" was taboo in the behaviourist epoch, consciousness is still not fully integrated into cognitive science. Perhaps 50–100 years from now, we will look back and wonder how we ignored something so fundamental.

Unconscious processes: Many functions happen unconsciously—walking, counting steps, breathing. We are not conscious of most low-level motor and perceptual processes.

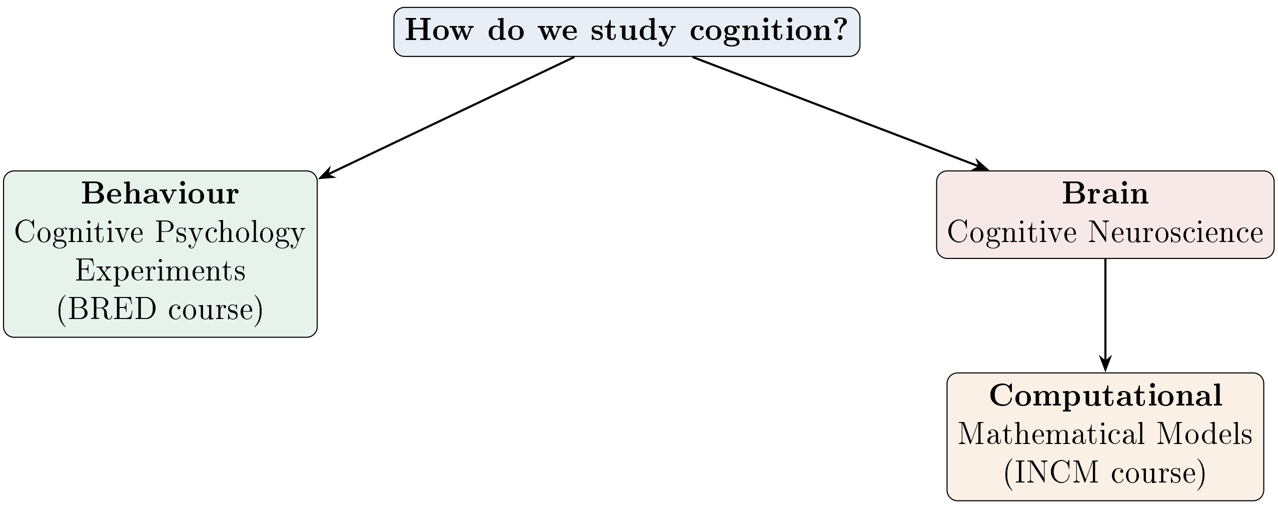
2.4 Brain: The Material Basis for Behaviour

Reductionist View

The **reductionist** position holds that everything we experience mentally has a **material basis in the brain**. Mind = brain activity. There is ongoing debate about whether consciousness can be fully reduced to neural activity.

Alternative view: Some traditions propose a non-material basis ("spirit" or "universal consciousness") that animates the body. This is difficult to test empirically. As scientists, we proceed with the assumption of material causation while acknowledging the open question.

2.5 How Do We Study Cognition?



Three pillars of investigation:

I. **Behaviour** — Cognitive psychology experiments (reaction time, accuracy)

II. **Brain** — Cognitive neuroscience (neuroimaging, patient studies)

III. **Computational Models** — Mathematical/computational models that bridge theory and experiment

2.5.1 Behavioural Experiments

- Conducted in controlled lab settings
- **Reaction time (RT)** tasks: present a stimulus, measure time to button press
- Average human reaction time ≈ **100 ms** (visual stimulus to motor response)
- Cricket batsmen need ∼15–50 ms response time—far below baseline RT; achieved through extensive training and anticipation
- Physical limitation: neural signal must travel from retina → visual cortex → motor cortex → spinal cord → muscles (electrochemical propagation is slow)

Additional tools: eye trackers, VR-based environments, physiological measurements (heart rate, skin conductance).

2.5.2 Studying the Brain: Healthy Participants

Non-invasive techniques:

- **EEG** (Electroencephalography) — electrodes on scalp measure electrical activity
- **MEG** (Magnetoencephalography) — measures magnetic fields from neural activity
- **fMRI** (functional MRI) — measures blood oxygenation changes (BOLD signal)

Invasive techniques:

- **PET** (Positron Emission Tomography) — inject radioactive tracer; measure glucose metabolism
- **ECoG** (Electrocorticography) — electrodes placed directly on the cortex (during surgery)

2.5.3 Studying the Brain: Patients and Neurological Disorders

Learning from Brain Damage

Much of what we know about brain–behaviour relationships comes from studying patients with brain injuries or neurodegenerative diseases. If area X is damaged and function Y is lost, we infer that area X participates in function Y.

Phineas Gage (1848)

A railroad worker had an iron rod shot through his **frontal lobe** (orbitofrontal cortex). He survived and was conscious, but his personality changed dramatically: he became impulsive, angry, socially inappropriate, and unable to plan. He lost his job and could not hold another.
Inference: The **frontal lobe** is critical for planning, social behaviour, personality, and emotional regulation—not vital for basic survival functions (breathing, heart rate) but essential for what makes us "human."

Broca's Aphasia

Paul Broca's patient could understand speech (follow commands like "pour water into the glass") but could *not produce* speech.
Affected area: Broca's area in the **frontal lobe**.
Type: Expressive (non-fluent) aphasia.
Key inference: Speech production and speech comprehension are *independent modules* served by different brain regions.

Wernicke's Aphasia

Carl Wernicke's patient could speak fluently but produced nonsensical speech and could *not comprehend* language.
Affected area: Wernicke's area in the **temporal lobe**.
Type: Receptive (fluent) aphasia.
Key inference: Confirms the **dissociation** between speech production (frontal) and comprehension (temporal).

Capgras Syndrome

Patients *recognise* their loved ones visually ("he looks like my husband, sounds like my husband") but *believe* they are impostors. They claim the real person has been abducted and replaced.

Analysis — What is intact:

- Visual recognition (pattern matching) — intact
- Memory retrieval and comparison — intact

What is lost:

- **Emotional evaluation** — the "warm feeling" or emotional connection to the recognised face

Inference — Two systems:

1. A system for pattern processing, memory retrieval, and comparison
2. A separate system for **emotional valuation** (involving the **amygdala**)

These two systems are integrated in the **frontal cortex**. Damage to the amygdala, the emotional pathway, or the integrating region causes Capgras syndrome.

Other conditions studied:

- **Alzheimer's dementia** — starts in temporal lobe; manifests as memory loss, disorientation
- **Frontotemporal dementia** — personality changes, impulsive behaviour, speech difficulties
- **Lobotomy** — historical (discredited) practice of removing frontal lobe tissue to treat personality disorders

2.5.4 The Role of Computational Modelling

Why Models Are Essential

1. **Manipulation** — in a model, you can switch components on/off and observe effects (impossible in vivo without waiting for specific diseases)
2. **Integration** — models combine observations from behaviour, neuroimaging, and patient studies into a unified framework
3. **Theory formation** — a complex system *cannot* be understood without a formal theory. No matter how sophisticated your measurements, without a model predicting how measurements relate, data remains uninterpretable.
4. **Theory–Experiment cycle** — model → prediction → experiment → refine model → new prediction → ...

This is exactly how physics works: cosmology does not work by observation alone; it requires theory.

Practice & Review

Q1. [Multiple Choice] The ability to focus on one conversation in a noisy party while ignoring all others is called:

(A) The cocktail party effect (B) Divided attention (C) Qualia (D) Dual-task processing

Answer: **A.** The cocktail party effect is a classic demonstration of *selective (focused) attention*—the ability to prioritise one stimulus stream and suppress others.

Q2. [Short Answer] Define qualia. Why does their existence make consciousness scientifically challenging to study?

Answer: Qualia are the subjective, personal qualities of conscious experiences—e.g., the precise "crunchiness" and "sweetness" *you* experience biting an apple. Your qualia are not identical to anyone else's and cannot be directly observed or verified by a third party. Because consciousness is inherently first-person and qualia cannot be measured objectively, standard scientific methods (which require repeatability and third-party verification) cannot easily access them.

Q3. [Essay] Compare EEG, fMRI, and PET as tools for studying the brain. For each, state what it measures, whether it is invasive, and its temporal/spatial resolution trade-off.

Answer: **EEG**—measures aggregate electrical activity of neurons via scalp electrodes (non-invasive); excellent *temporal* resolution (∼ms) but poor spatial resolution. **fMRI**—measures the BOLD signal (blood oxygenation level-dependent, an indirect proxy for neural activity, non-invasive); excellent *spatial* resolution (mm) but poor temporal resolution (∼seconds). **PET**—measures glucose metabolism via injected radioactive tracer (invasive); poor temporal resolution (minutes); largely replaced by fMRI for most research since ∼2000. Trade-off summary: EEG = when + what happened; fMRI/PET = where it happened.

Q4. [Short Answer] Describe the Capgras syndrome case. What two dissociated brain systems does it reveal?

Answer: Capgras patients visually recognise their loved ones ("He looks and sounds exactly like my husband") yet believe them to be imposters—they insist the real person has been replaced. What is *intact*: (1) visual pattern recognition and memory comparison. What is *lost*: (2) emotional valuation—the "warm feeling" of familiarity (mediated by the amygdala and its connection to the frontal cortex). This reveals two independent systems: a cognitive/recognition system and a separate affective/emotional evaluation system, normally integrated in the frontal cortex.

Q5. [Multiple Choice] Broca's area damage causes a patient to:

(A) Speak fluently but produce nonsensical language (B) Understand speech but be unable to produce speech
(C) Be unable to understand speech but produce fluent output (D) Lose all language ability entirely

Answer: **B.** Broca's aphasia (expressive/non-fluent aphasia) results from damage to Broca's area in the frontal lobe. Speech comprehension (Wernicke's area, temporal lobe) remains intact.

Q6. [Short Answer] What happened to Phineas Gage and what did it demonstrate about the frontal lobe?

Answer: In 1848, a railroad worker survived an iron rod blasting through his orbitofrontal cortex. Although physically functional, his personality changed dramatically: he became impulsive, irritable, unreliable, socially inappropriate, and unable to plan or hold a job. His survival functions (breathing, heart rate) were unaffected. The case demonstrated that the frontal lobe is critical for **executive function** (planning, inhibitory control), **social behaviour**, and **personality**—the very functions that define us as human.

Q7. [Short Answer] Name the three pillars of cognitive science investigation introduced in this lecture. Which uses the BOLD signal?

Answer: (I) **Behaviour**—cognitive psychology experiments (reaction time, accuracy); (II) **Brain**—cognitive neuroscience (neuroimaging, patient studies); (III) **Computational models**—mathematical models linking theory to experiment. The BOLD (Blood Oxygenation Level-Dependent) signal is used by **fMRI** in pillar II.

Q8. [Essay] Why is computational modelling considered essential in cognitive neuroscience, not merely supplementary?

Answer: A complex system cannot be understood from measurement alone—a formal theory is required to *predict* how measurements should relate to each other. Modelling provides: (1) **Manipulation**—in a model you can selectively disable components (impossible in a living brain without waiting for specific disease); (2) **Integration**—models synthesise findings from behaviour, imaging, and patient studies into a unified framework; (3) **Theory formation**—generates testable predictions; (4) **Theory–experiment cycle**—model $\to$ prediction $\to$ experiment $\to$ model refinement. Analogous to how physics could not advance without mathematical theory (e.g., cosmology without General Relativity).

Lecture 3: Neuroanatomy — Brain Structure (March 10)

3.1 Overview of the Brain

The brain is:

- Divided into **two hemispheres** (left and right), connected by dense fibres (e.g., **corpus callosum**)
- Composed of the **cerebrum**, **cerebellum**, and **brainstem/spinal cord**
- The cerebrum is also called the **neocortex** ("new cortex")
- Protected by the **skull**, surrounded by **meninges** (including pia mater)

3.2 Grey Matter and White Matter

Grey Matter

Grey matter consists of neuronal **cell bodies** (soma). It appears darker. Found on the *outer surface* of the cerebral cortex and in the *interior* of the spinal cord.

White Matter

White matter consists of **myelinated axonal fibres** that connect neurons across regions. The myelin sheath (a fatty coating) gives it its white appearance. Found *inside* the cerebrum and on the *outside* of the spinal cord.

Key Contrast: Cortex vs. Spinal Cord

- **Cortex:** grey matter (outside), white matter (inside)
- **Spinal cord:** grey matter (inside, butterfly-shaped), white matter (outside)

This is an *opposite* arrangement!

An MRI scan reveals four tissue types: grey matter, white matter, cerebrospinal fluid (CSF), and bone.

3.3 Cerebrospinal Fluid (CSF)

Cerebrospinal Fluid

CSF is a clear, colourless fluid that surrounds the brain and spinal cord. It:

- Provides cushioning and mechanical protection
- Supplies nutrients
- Removes waste products
- Circulates through ventricles and channels within the brain

3.4 Central and Peripheral Nervous Systems

Central Nervous System (CNS)

The **CNS** comprises the **brain** (cerebrum + cerebellum) and the **spinal cord**. It receives, processes, and integrates information, and generates output commands.

Peripheral Nervous System (PNS)

The **PNS** consists of all the nerves outside the CNS. It includes:

- **Sensory neurons** — collect information from receptors (touch, vision, hearing, smell, taste) and send it to the CNS
- **Motor neurons** — transmit commands from the CNS to muscles and glands

Sensory–Motor loop:

$$\text{Receptors(PNS)} \xrightarrow{\text{sensory signals}} \text{CNS(Brain + Spinal Cord)} \xrightarrow{\text{motor commands}} \text{Muscles(PNS)}$$

Example: Touch receptors on the skin sense mechanical distortion → electrical signals travel via axons to spinal cord → processed in brain → motor commands sent down spinal cord → muscles contract/relax (e.g., biceps contracts + triceps relaxes → arm bends at elbow).

3.5 Brain Sections (Planes of Cut)

Section	Plane	Shows
Coronal	Front-to-back cut	Both hemispheres (L & R)
Horizontal (Transverse/Axial)	Top-to-bottom cut	Both hemispheres (L & R)
Mid-Sagittal	Left-right midline cut	One hemisphere only

Directional terminology:

- **Anterior** (front) / **Posterior** (back)
- **Superior/Rostral** (top) / **Inferior/Caudal** (bottom)
- **Ventral** (front side in humans) / **Dorsal** (back side in humans)
- (Note: these terms originate from quadrupedal animals, causing some confusing mapping to bipedal humans)

3.6 Cortical Surface Features

Gyrus, Sulcus, Fissure

- **Gyrus** (pl. gyri) — a "bump" or ridge on the cortical surface
- **Sulcus** (pl. sulci) — a "valley" or groove between gyri
- **Fissure** — a deep, prominent groove (larger than a sulcus)

The cortex is folded (convoluted) to pack more surface area into the skull.

Major landmarks:

Landmark	Separates
Central Sulcus	Frontal lobe (anterior) from Parietal lobe (posterior)
Lateral (Sylvian) Fissure	Temporal lobe (inferior) from Frontal/Parietal (superior)
Longitudinal Fissure	Left hemisphere from Right hemisphere
Transverse Fissure	Cerebrum from Cerebellum

3.7 The Four Lobes of the Cerebrum

3.7.1 (1) Frontal Lobe

Located **anterior** to the central sulcus. The **newest** part of the neocortex; most developed in humans.

Frontal Lobe Functions

- **Executive function**: working memory, planning, decision-making, inhibitory control
- **Motor cortex**: strip just anterior to central sulcus; controls all voluntary motor actions
- **Speech production**: Broca's area (in frontal lobe)
- **Emotional evaluation**, social behaviour, personality
- Continues developing until age ∼**21 years**

Executive function includes:

- **Working memory** — temporary scratch-pad for ongoing tasks
- **Inhibitory control** — "go/no-go"; knowing when *not* to act (develops in children; why infants put fingers in electrical sockets)
- **Planning** — sequencing actions (e.g., making a shopping list → take money → take bag → go to market → buy items); long-horizon planning (career, retirement)

"Out of Sight, Out of Mind" — Working Memory Development

Object permanence experiment: A toy is shown to an infant and moved behind a screen.

- A ∼5-month infant: loses interest once the toy goes behind the screen (no working memory persistence)
- A ∼6-month infant: tracks and expects the toy to emerge on the other side

Working memory capacity increases with age, reaching adult levels around 8–9 years.

Phineas Gage revisited: Damage to frontal lobe → loss of planning, social behaviour, personality change—but basic survival functions remained intact. Frontal lobe is "not vital for breathing but essential for being human."

3.7.2 (2) Parietal Lobe

Located **posterior** to the central sulcus.

Parietal Lobe Functions

- **Somatosensory cortex**: strip just posterior to central sulcus; receives sensation from all over the body
- **Proprioception**: sense of where body parts are in space without looking (close eyes → know your hand is raised)
- **Spatial orientation**: ego-centric (relative to self) and allocentric/world-centred (relative to other objects)
- **Goal-directed action**: guides motor actions to spatial targets (e.g., swatting a mosquito on your arm)

Vestibular system: Sensors in the inner ear detect orientation/balance. Ear infections can cause vertigo ("world is moving") because the sensor provides incorrect orientation data.

3.7.3 (3) Occipital Lobe

Located at the **back** of the brain, above the transverse fissure.

Occipital Lobe Functions

- **Visual cortex**: primary centre for all visual processing
- Object recognition and identification
- Colour, shape, and size perception
- Object motion detection (direction, velocity)
- Two visual streams: one for **object recognition** ("what"), one for **motion/spatial** ("where/how")

3.7.4 (4) Temporal Lobe

Located beneath the lateral (Sylvian) fissure, under the temporal bone.

Temporal Lobe Functions

- **Auditory cortex**: processing all sounds
- **Speech comprehension**: Wernicke's area (in temporal lobe)
- **Language processing**
- **Object naming**: associating labels ("chair", "table") with recognised objects
- Input from ears → hair cells convert pressure waves to electrical signals → auditory cortex

3.7.5 (5) Insula ("Hidden" Fifth Lobe)

Tucked inside the temporal lobe, not visible from the outside.

- **Sense of self**
- **Saliency processing** — which stimuli are more/less salient

3.8 Subcortical Structures

Clusters of neurons located *beneath* the cortex. These are phylogenetically **older** structures, present even in simpler animals.

Structure	Function
Amygdala	Fear processing; emotional response to threatening stimuli
Hippocampus	Long-term memory formation (horseshoe-shaped)
Basal Ganglia	Motor control; gateway for motor output to cortex
Thalamus	Sensory relay station; gathers incoming sensory info & distributes to cortex
Hypothalamus	Hunger, thirst, temperature regulation, breathing, rage
Cerebellum	Motor coordination; balance; motor learning

Thalamus and Basal Ganglia as Gateways

Thalamus = gateway for *sensory* information **into** the cortex.
Basal Ganglia = gateway for *motor* commands **out of** the cortex.

Cerebellum and Alcohol

Long-term alcohol abuse **compromises the cerebellum**, leading to loss of motor coordination (swaying, inability to walk straight). This is also why police use the "walk in a straight line" test for intoxication—alcohol temporarily shuts down cerebellar signals. The condition is called **cerebellar ataxia**.

3.9 The Midbrain and Brainstem (Old Brain)

The **midbrain/brainstem** controls the most vital functions:

- Breathing, respiration
- Heart rate regulation
- Sleep–wake cycle
- Temperature control

If You Had to Lose Part of Your Brain...

Given a choice, one would rather lose the **cerebrum** (neocortex) than the **midbrain**. The midbrain controls survival-critical functions. Many motorcycle accident victims enter **coma** because the brainstem (where breathing circuits reside) is damaged by mechanical shock.
Wear a helmet—it lessens shock to the brain, especially the brainstem–spinal cord junction.

3.10 Brain Development

- The CNS is **not** fully developed at birth
- Full maturation continues until approximately **age 21**
- Developmental milestones: visual system set by ~1 year, auditory/speech by ~1.5–2 years, frontal lobe by ~18–21 years
- Development involves *growing connections* (not growing new brain tissue)—inactive connections become active through experience
- Humans have the **longest post-birth gestation period** of any animal: rolling → raising head → crawling → standing → walking → speech → sense of self

Mirror Test for Self-Awareness

A chimpanzee with a dot on its forehead was placed in front of a mirror. It tried to wipe the dot *off the mirror* rather than off its own forehead—indicating a lack of self-recognition. In human infants, the sense of self emerges gradually during development.

Practice & Review

Q1. [Multiple Choice] Grey matter in the brain primarily consists of:

(A) Myelinated axonal fibres (B) Cerebrospinal fluid (C) Neuron cell bodies (soma) (D) Synaptic vesicles

Answer: **C.** Grey matter is composed of neuronal cell bodies and their supporting glial cells. It appears darker in colour. White matter consists of myelinated axons (myelin gives the white appearance).

Q2. [Short Answer] What is the key difference in the arrangement of grey and white matter between the cerebral cortex and the spinal cord?

Answer: In the **cerebral cortex**: grey matter is on the *outside* (cortical surface), white matter is on the *inside*. In the **spinal cord**: the arrangement is *reversed*—grey matter forms a butterfly-shaped core on the *inside*; white matter surrounds it on the *outside*. This opposite arrangement is a frequently examined fact.

Q3. [Short Answer] Name the four lobes of the cerebrum and give one primary function of each.

Answer: (1) **Frontal lobe**—executive function, motor control (motor cortex), speech production (Broca's area), planning, personality; (2) **Parietal lobe**—somatosensory processing, proprioception, spatial orientation; (3) **Occipital lobe**—visual processing (primary visual cortex); (4) **Temporal lobe**—auditory processing, speech comprehension (Wernicke's area), object naming.

Q4. [Multiple Choice] The groove (valley) between two gyri on the cortical surface is called a:

(A) Sulcus (B) Gyrus (C) Fissure (D) Ventricle

Answer: **A.** A *sulcus* is a groove; a *gyrus* is a ridge/bump. A *fissure* is a deep, major groove (e.g., the lateral/Sylvian fissure separating the temporal lobe from the frontal/parietal lobes).

Q5. [Essay] Describe the frontal lobe's role in executive function. What are its three components? Use the Phineas Gage case and the "object permanence" experiment to illustrate its development.

Answer: Executive function has three components: (1) **Working memory**—temporary scratch-pad for ongoing tasks; (2) **Inhibitory control**—"go/no-go" responses; knowing when *not* to act (reason infants put fingers in sockets—frontal lobe not yet mature); (3) **Planning**—sequencing actions over time (shopping list, career planning). **Phineas Gage**: iron rod through orbitofrontal cortex →

lost planning, social behaviour, and personality while survival functions remained intact—frontal lobe is "not vital for breathing but essential for being human." **Object permanence**: a ∼5-month infant loses interest when a toy disappears behind a screen (no working memory persistence); a ∼6-month infant tracks and waits for it (working memory emerging). Frontal lobe matures until ∼21 years.

Q6. [Short Answer] Contrast the roles of the thalamus and the basal ganglia. Use the "gateway" metaphor.

Answer: **Thalamus** = *sensory gateway into the cortex*—collects all incoming sensory information and distributes it to the appropriate cortical areas. **Basal ganglia** = *motor gateway out of the cortex*—a relay for motor output commands. They operate in opposite directions: thalamus (in), basal ganglia (out).

Q7. [Multiple Choice] Long-term alcohol abuse damages the cerebellum, leading to:

(A) Memory loss (Korsakoff syndrome) (B) Cerebellar ataxia (loss of motor coordination)
(C) Inability to speak (D) Complete personality change

Answer: **B.** Chronic alcohol compromises the cerebellum, causing *cerebellar ataxia*—swaying gait, inability to walk in a straight line, loss of balance and coordination. This is the basis of the police "walk the line" sobriety test.

Q8. [Short Answer] List three functions of cerebrospinal fluid (CSF) and describe where it is found.

Answer: CSF is a clear, colourless fluid that surrounds the brain and spinal cord (circulating through ventricles and channels). Functions: (1) **Mechanical cushioning**—protects the brain from physical impact; (2) **Nutrient delivery**—supplies oxygen and nutrients to CNS tissue; (3) **Waste removal**—clears metabolic by-products; (4) **Pressure regulation**—maintains stable intracranial pressure.

Q9. [Short Answer] What is proprioception and which lobe handles it? Give an example of proprioception in action.

Answer: Proprioception is the sense of the position and orientation of one's own body parts in space *without looking*. It is processed in the **parietal lobe** (somatosensory cortex). Example: close your eyes and raise your right hand—you know where your hand is without visual feedback. Proprioceptive signals come from receptors in muscles, tendons, and joints.

Lecture 4: Brain Imaging, Neurons, and Scales (March 13)

4.1 Brain Evolution: The Triune Brain

The **triune brain theory** (Paul MacLean) proposes three evolutionary layers:

Neomammalian (Neocortex) — higher-order: reasoning, planning, language

↑

Paleomammalian (Limbic System) — basic emotions: happy, sad, fear

↑

Reptilian Complex — basic survival: respiration, heart rate, temperature

- The older layers are **co-opted** from evolutionary ancestors (shared with reptiles, lower mammals)
- The **prefrontal cortex** is the most developed in humans—supports planning, decision-making, emotional regulation, higher-order thinking
- Basic functions (respiration, hunger, temperature) are common across species
- Studying reptiles and mammals helps us understand our own brain
- Evolutionary trade-offs: e.g., dogs have superior smell but inferior visual processing compared to humans

4.2 Resting State Brain Function

Resting State

Resting state refers to brain activity when a person is *not* performing any specific task (e.g., lying in an MRI scanner with eyes closed). The brain remains highly active even at "rest."

- Resting state brain connectivity differs between healthy individuals and those with:
  - Alzheimer's disease
  - Dementia
  - Coma / vegetative state
- This has become a powerful diagnostic tool for neurological and psychiatric conditions

- **Application:** predicting recovery in coma patients from resting-state fMRI patterns (still an open research problem)
- The **brain operates 24/7/365**—if it stops, you are dead

4.3 Brain Imaging Techniques (Detailed)

4.3.1 Positron Emission Tomography (PET)

PET Scan

PET measures **glucose metabolism** in the brain. A radioactive tracer (radioisotope) is injected into the bloodstream, travels to the brain, and as it decays, emits positrons that are detected by the scanner.

- Active brain regions consume more glucose → higher metabolism → "hot" colours (red, yellow)
- **Example:** Eyes closed (low occipital activity) → Eyes open (occipital cortex activates) → Complex visual scene (further increased visual cortex activity)
- **Invasive**: requires injection of radioactive substance
- Older technique (1980s–90s); less commonly used today

4.3.2 Structural MRI

- Uses magnetic resonance to image brain *structure*
- Shows grey matter, white matter, CSF, and bone with different contrasts
- Used to identify structural damage (tumours, lesions, atrophy)
- Three views: coronal, horizontal/axial, sagittal

4.3.3 Electroencephalography (EEG)

EEG

EEG measures the aggregate **electrical activity** of neurons from electrodes placed on the scalp. It captures the synchronised firing of ~10,000+ neurons in a region.

EEG frequency bands and brain states:

State	Frequency	Amplitude
Deep sleep	Low (delta band)	High amplitude
Drowsy (pre-sleep)	Medium (alpha band)	Medium
Awake & alert	High (beta/gamma band)	Low amplitude

EEG Signatures

- **Awake:** high frequency, low amplitude — neurons firing independently and rapidly, exchanging information
- **Deep sleep:** low frequency, high amplitude — neurons firing synchronously in slow waves
- Frequency bands: **delta** (lowest) < **theta** < **alpha** < **beta** < **gamma** (highest)

Event-Related Potentials (ERPs):

- Present a stimulus (e.g., auditory tone) and measure the brain's electrical response
- Time-locked to stimulus onset; averaged over many trials
- Signal amplitude: $\sim$2 μV; duration: $\sim$700 ms post-stimulus
- ERP components reveal stages of information processing

EEG systems: 8, 16, 32, 64, 128, or 256 electrodes. More electrodes = denser spatial sampling. Electrodes arranged using the **10–20 system** (every 20% of head distance).

MEG (Magnetoencephalography): Measures the *magnetic field* generated by the same electrical activity that EEG detects. Requires very sensitive sensors (SQUIDs).

4.3.4 Functional MRI (fMRI)

fMRI

fMRI measures the **Blood Oxygenation Level-Dependent (BOLD)** signal. When a brain region is active, it demands more oxygenated blood. The ratio of oxygenated to deoxygenated haemoglobin changes the local magnetic signal, which the MRI detects.

PET vs. fMRI

- **PET:** measures glucose *metabolism*; invasive (radioactive tracer)
- **fMRI:** measures blood *oxygenation* changes; non-invasive
- Both are *indirect* measures of neural activity
- fMRI breakthrough: $\sim$1999–2000; transformed cognitive neuroscience by allowing functional studies without invasive procedures

Key fMRI finding — Mental imagery:

- When participants *view* pictures: strong activity in visual cortex (occipital lobe)
- When participants *imagine* objects (no physical stimulus): visual cortex still activates, but with *reduced* intensity

- Higher-order visual areas show similar activity for real and imagined stimuli
- Same result for motor imagery: imagining a hand movement activates motor cortex (less intensely than actual movement)

4.4 Brain–Computer Interfaces (BCIs)

Brain–Computer Interface

A **BCI** decodes brain activity (e.g., motor intention from EEG/ECoG) and interfaces it with a computer or robotic device, enabling control through thought alone.

Application: Patients with paralysis (e.g., stroke, ALS/motor neuron disease like Stephen Hawking) who can still *think* about movement can have their motor intentions decoded and used to control a robotic arm, generate speech, or type.

4.5 Deep Brain Stimulation (DBS)

Deep Brain Stimulation

DBS involves surgically implanting electrodes into specific subcortical structures (e.g., basal ganglia) and delivering a mild electrical current to modulate neural activity.

- Primary application: **Parkinson's disease** (a motor disorder with tremors and rigidity)
- When DBS is switched **on**: symptoms dramatically reduce; patient can move normally
- When DBS is switched **off**: symptoms gradually return
- Used when medication is insufficient in severe cases

4.6 The Neuron

Neuron

A **neuron** is the fundamental computational unit of the nervous system. It is an **electrically excitable** cell with three main parts:

1. **Dendrites** — receive input from other neurons
2. **Cell body (soma)** — aggregates inputs spatially and temporally
3. **Axon** — generates and propagates the output (**action potential**) to other neurons

Key properties:

- Action potential amplitude: ∼30 mV (from resting ∼–70 mV)
- Average connectivity: each neuron connects to $\sim 10^4$ (10,000) other neurons
- Signal propagation is **regenerative**: the action potential is regenerated at full amplitude at each node, not split/attenuated (unlike an electrical wire splitting current among branches)
- The **myelin sheath** (fatty coating on axons) gives rise to white matter

4.7 Synapses

Synapse

A **synapse** is the junction between an axon terminal and the dendrite of the next neuron. The gap (∼1 μm) is called the **synaptic cleft**. Communication is **electro-chemical**:

1. Electrical signal (action potential) arrives at axon terminal
2. Chemical **neurotransmitters** are released into the synaptic cleft
3. Neurotransmitters bind to receptors on the post-synaptic dendrite
4. A new electrical signal is generated in the receiving neuron

This electrochemical process makes neural communication **slow** compared to electronic circuits.

4.8 Spatial Scales of the Nervous System

Scale	Size	Structure
Molecular	Ångströms (10^{-10} m)	Ion channels, neurotransmitters
Synaptic	Micrometres (μm)	Synaptic cleft, vesicles
Neuronal	∼100 μm	Single neuron cell body
Networks	Millimetres	Local circuits of connected neurons
Maps	Centimetres	Motor map, somatosensory map
Systems	∼10 cm	Visual system, auditory system
CNS	∼1 m	Entire brain + spinal cord

4.9 Temporal Scales of Cognitive Phenomena

Time Scale	Order	Phenomenon
10^{-3} s (1 ms)	Milliseconds	Action potential firing
10^{-1} s (100 ms)	Hundred ms	Object recognition; simple reaction time
10^{0} s (1 s)	Seconds	Simple motor action
10^{1} s (10 s)	Tens of seconds	Habituation
10^{2} s (100 s)	Minutes	Planning a chess move
10^{5}–10^{6} s	Days–Months	Perceptual learning; skill acquisition (e.g., cycling)
10^{7} s	Months–Years	Learning to walk (infant)

4.10 Memory and Learning at the Synapse

Synaptic Plasticity

Learning occurs by **increasing the strength of synaptic connections**. All memory is encoded in connection strengths. In some cases, previously inactive connections become active through repeated experience.
This is the neural basis of the **nature vs. nurture** debate: the "wiring" is in place (nature), but connections are refined through experience (nurture). Both contribute to cognitive development.
Nobel Prize (2000): Awarded for discovery of synaptic changes due to learning.
Nobel Prize (1957): Awarded for discovery of action potential mechanisms.

4.11 Brain vs. Digital Computer

Feature	**Digital Computer**	**Brain**
Processing unit	Complex CPU (ALU); GHz speed	Simple neuron; ms speed
Number of processors	1–thousands	$\sim 10^{10}$ (billions)
Processing style	Sequential (mostly)	Massively **parallel**
Speed	$\sim 10^9$ operations/s	∼ms per operation (slow)
Communication	Electrical (fast)	Electrochemical (slow)
Memory architecture	Separate (RAM, HDD, cache); localised	**Distributed**; memory at every synapse
Memory addressing	Address-based (location)	**Content-addressable**
Learning	Stored program	**Self-learning**
Robustness	Vulnerable (heat, damage)	**Robust** (works at 38°C+; tolerates injury)
Reliability	100% deterministic	Probabilistic; graceful degradation
Data type	Numerical & symbolic	Conceptual
Operating environment	Well-defined, constrained	Unconstrained, real-world

Content-Addressable Memory

In a computer, memory is accessed by **address** (location). In the brain, memory is **content-addressable**: you recall your friend's face by thinking about them, not by sending an address to a specific memory location. The memory is distributed across synapses throughout the brain, not stored in one place.

Practice & Review

Q1. [Multiple Choice] fMRI primarily measures which of the following to infer brain activity?

(A) Electrical impulses on the scalp (B) Changes in blood oxygenation (BOLD signal)
(C) Glucose metabolism via radiotracer (D) Magnetic fields from ion channels

Answer: **B.** fMRI detects the Blood Oxygenation Level-Dependent (BOLD) signal. Active brain regions demand more oxygenated blood; the ratio of oxy- to deoxy-haemoglobin changes the local MRI signal. It is non-invasive and replaced PET as the dominant functional imaging method after ∼2000.

Q2. [Short Answer] Describe the triune brain theory. Name the three layers from oldest to newest and identify which part is most developed in humans.

Answer: The triune brain (Paul MacLean) proposes three evolutionary layers: (1) **Reptilian complex** (brainstem)—basic survival: respiration, heart rate, temperature; (2) **Paleomammalian brain** (limbic system)—basic emotions: fear, happiness, anger; (3) **Neomammalian brain** (neocortex)—higher-order reasoning, planning, language. The **prefrontal cortex** (part of the neocortex) is most developed in humans and supports our most advanced cognitive capacities.

Q3. [Essay] Explain brain–computer interfaces (BCIs). Describe two types, their invasiveness, and a clinical application for each.

Answer: A BCI decodes brain activity (motor intention, thoughts) and interfaces it with a computer or robotic device, enabling control through thought alone. (1) **Surface BCI** (non-invasive): EEG electrodes on the scalp decode motor intent. Application: IIIT research decoding motor intention from EEG for stroke/hemiplegia patients to drive physiotherapy robots. Limitation: lower signal-to-noise. (2) **Implanted BCI** (invasive): electrodes implanted in brain tissue (e.g., Neuralink). Application: enabling paralysed patients (ALS, spinal cord injury) to control robotic arms, type, or communicate. Higher precision but requires surgery with associated infection risk. Both types demonstrate the power of understanding motor intention circuitry.

Q4. [Short Answer] What is synaptic plasticity? Connect it to memory and to the nature vs. nurture debate.

Answer: Synaptic plasticity is the ability of synaptic connections to strengthen or weaken based on activity patterns (e.g., Hebbian learning: "neurons that fire together, wire together"). All memory is encoded in synaptic connection strengths—learning = selectively strengthening relevant connections through repeated experience (Nobel Prize 2000). Nature–nurture connection: the neural "wiring" (number of neurons, basic connectivity) is genetically determined (nature), but which specific connections become strong or weak depends on experience (nurture). Both contribute.

Q5. [Multiple Choice] In EEG recordings, a state of deep sleep is characterised by:

(A) High frequency, low amplitude activity (B) Low frequency, high amplitude activity
(C) Complete electrical silence (D) High frequency, high amplitude activity

Answer: **B.** Deep sleep = delta band (lowest frequency, highest amplitude); neurons fire synchronously in slow waves. Being awake and alert = beta/gamma band (high frequency, low amplitude); neurons fire independently and rapidly, exchanging information. The inverse relationship of frequency and amplitude is a key EEG principle.

Q6. [Short Answer] Name the five EEG frequency bands from lowest to highest and give the associated brain state for each extreme.

Answer: **Delta** (0.5–4 Hz): deep sleep; **Theta** (4–8 Hz): drowsy, light sleep; **Alpha** (8–13 Hz): relaxed but awake, eyes closed; **Beta** (13–30 Hz): active, attentive; **Gamma** (>30 Hz): highly focused processing, problem-solving. The 10–20 electrode placement system spaces electrodes every 20% of head distance.

Q7. [Short Answer] Explain Deep Brain Stimulation (DBS) for Parkinson's disease. What structure is targeted and what is the evidence for its effectiveness?

Answer: DBS involves surgically implanting an electrode into the **subthalamic nucleus** (part of the basal ganglia) and delivering a mild electrical current. Parkinson's disease involves dysfunction of the basal ganglia leading to resting tremor and rigidity. Evidence: when DBS is switched **on**, the patient's tremor dramatically reduces within seconds and they can move normally; when switched **off**, tremors gradually return over minutes. The patient carries a switch. Used for advanced Parkinson's when medication is insufficient. DBS demonstrates the precise role of the subthalamic nucleus in motor control.

Q8. [Essay] Compare the brain with a digital computer across five dimensions. Which comparison is most surprising and why?

Answer: Key comparisons: (1) **Processing**: computer = sequential; brain = massively parallel ($\sim 10^{10}$ neurons). (2) **Speed**: computer = GHz (10^9 ops/s); brain = ms per operation (slow), but compensates with parallelism. (3) **Memory**: computer = address-based (location); brain = **content-addressable** (recalled by content, distributed across synapses). (4) **Learning**: computer = stored program; brain = self-learning (synaptic plasticity). (5) **Robustness**: computer = brittle (single bit flip can crash); brain = robust, works at 38 °C+, tolerates neuron death with graceful degradation. Most surprising: content-addressable memory—recalling your friend by thinking about them, without knowing a "memory address."

Lecture 5: Perception, Neural Coding, and Organisation Principles (March 17)

5.1 Perception

Perception

Perception is the interpretation of raw sensory input data. It is not passive recording but an active, constructive process by which the brain makes sense of incoming signals.

Visual illusions demonstrate that perception is not veridical (faithful to physical reality)—the brain constructs interpretations that can be "wrong" relative to the physical stimulus.

5.2 Neurons: Electrically Excitable Cells

Recap and extension:

- Neurons are **electrically excitable**—they respond to stimulation by generating electrical discharges (action potentials)
- **Axons carry information** from the cell body to distant targets
- Unlike other body cells, neurons have the unique property of electrical excitability

5.3 Synapses: Electrochemical Transmission

- The *majority* of synapses in the brain are **chemical synapses** (electrochemical transmission via neurotransmitters)
- A minority are **gap junctions** (direct electrical coupling), which have been **reduced evolutionarily**
- Chemical synapses are slower but allow for modulation, learning, and plasticity

5.4 Hodgkin–Huxley Model

Hodgkin–Huxley Model

The **Hodgkin–Huxley model** (Nobel Prize, 1963) is a mathematical model describing how action potentials are initiated and propagated in neurons. Developed using the **giant squid axon** as a model system.
Key voltage values:

- **Resting potential:** ≈ -70 mV
- **Threshold potential:** ≈ -55 mV (when exceeded, an action potential fires)

When the membrane potential depolarises from -70 mV past -55 mV, an **all-or-nothing** action potential is triggered ($\sim$+30 mV peak), propagates down the axon, and is regenerated at each node.

5.5 Inter-Pulse Intervals and Neural Coding

- Neurons encode information in the **timing** and **rate** of action potentials
- **Inter-pulse intervals** (time between successive spikes) carry information about stimulus intensity and other features
- Higher firing rates generally indicate stronger stimulation

5.6 From Action Potentials to Perception: Modularity

Modularity

Modularity is the principle that different cognitive functions are served by distinct, specialised brain modules (regions). The path from raw neural activity (action potentials) to rich perceptual experience involves hierarchical and modular processing.

5.7 Grandmother Coding vs. Distributed Processing

Grandmother Cell (Localist Theory)

The **grandmother cell hypothesis** proposes that specific individual neurons represent specific complex concepts (e.g., one neuron for your grandmother's face). This is a **localist** theory of representation.

Distributed Processing

Distributed (population) coding proposes that representations are spread across *many* neurons. No single neuron encodes a complete concept; instead, the pattern of activity across a population encodes the representation. This is also called **multi-modal distribution**.

Localist vs. Distributed

- **Grandmother cell:** one neuron = one concept. Simple but fragile (if that neuron dies, the concept is lost).
- **Distributed processing:** a pattern across many neurons = one concept. Robust (damage to some neurons causes graceful degradation, not total loss).
- Modern evidence supports **distributed processing** as the dominant coding scheme, though some neurons do show remarkable selectivity (e.g., "Jennifer Aniston neurons" in the medial temporal lobe).

Practice & Review

Q1. [Multiple Choice] The primary distinction between sensation and perception is:

(A) Sensation is the processing and transmission of sensory input; perception is the brain's *interpretation* of that input
(B) Sensation is conscious; perception is unconscious
(C) Sensation occurs in the brain; perception in the peripheral nervous system
(D) They are synonymous terms for the same process

Answer: **A.** Sensation happens at the sensory organs and is conveyed to the CNS; perception is the constructive, interpretive process in the brain that creates our internal model of reality. Perception is not a faithful copy of the world—it is an active reconstruction (demonstrated by visual illusions).

Q2. [Short Answer] What is the Hodgkin–Huxley model? State the two critical voltage values and what happens at each.

Answer: The Hodgkin–Huxley model (Nobel Prize 1963, developed using the giant squid axon) mathematically describes how action potentials are initiated and propagated. (1) **Resting potential** ≈ -70 mV: the stable negative charge across the neuron membrane when at rest. (2) **Threshold potential** ≈ -55 mV: when depolarisation exceeds this, an *all-or-nothing* action potential fires, reaching $\approx +30$ mV peak, and propagates regeneratively down the axon.

Q3. [Essay] Compare the grandmother cell (localist) theory with distributed (population) coding. What does the evidence support? What are the implications for brain robustness?

Answer: **Grandmother cell hypothesis**: one specific neuron encodes one specific complex concept (e.g., your grandmother's face). Simple but fragile: if that neuron dies, the concept is irretrievably lost. **Distributed (population) coding**: a concept is encoded as a *pattern of activity* across many neurons; no single neuron holds the complete representation. Robust: losing a few neurons causes *graceful degradation* (partial loss of fidelity, not total loss). Modern evidence strongly supports distributed coding as the dominant scheme. However, highly selective neurons exist—"Jennifer Aniston neurons" in the medial temporal lobe respond almost exclusively to a specific person's face/name—suggesting a continuum rather than a binary choice. Robustness implication: the brain can tolerate neuronal death (e.g., stroke, Alzheimer's progression) with gradual decline rather than catastrophic failure.

Q4. [Multiple Choice] The Kanizsa triangle illusion (a white triangle perceived where none is drawn) demonstrates:

(A) Top-down processing using prior knowledge (B) Bottom-up processing driven by stimulus properties
(C) Working memory capacity limits (D) Classical conditioning

Answer: **B.** Bottom-up processing: the visual system constructs contours from the arrangement of pac-man shapes, regardless of the viewer's language or cultural knowledge. This is not driven by prior expectations.

Q5. [Short Answer] Explain the difference between top-down and bottom-up perceptual processing. Give one specific example of each from the lecture.

Answer: **Bottom-up**: stimulus properties directly drive perception without prior knowledge. Example: Kanizsa triangle (all viewers, regardless of language, perceive the triangle); café wall illusion (parallel lines appear sloped due to contrast). **Top-down**: stored knowledge, templates, and expectations guide interpretation of ambiguous stimuli. Example: reading a word printed with occluded letters (stripes over "IBC")—your knowledge of English fills the gaps. A person who does not know English cannot read the occluded word.

Q6. [Short Answer] What is modularity in the context of brain organisation? Why is the modular arrangement efficient?

Answer: Modularity: different cognitive functions (vision, audition, motor control, language) are served by distinct, specialised brain regions (modules). Efficient because: (1) modules can process in *parallel*; (2) damage to one module selectively impairs that function without destroying others (selective deficits, e.g., Broca's vs. Wernicke's aphasia); (3) evolutionary refinement of one module does not require redesigning the whole system.

Q7. [Multiple Choice] The majority of synaptic connections in the brain use which type of transmission?

(A) Direct electrical (gap junctions) (B) Electrochemical (chemical synapses with neurotransmitters)
(C) Optical (photon-based) (D) Mechanical (pressure-based)

Answer: **B.** Chemical synapses are the dominant form; gap junctions have been *reduced* evolutionarily. Chemical synapses are slower but allow for modulation, learning (via neurotransmitter release modulation), and synaptic plasticity.

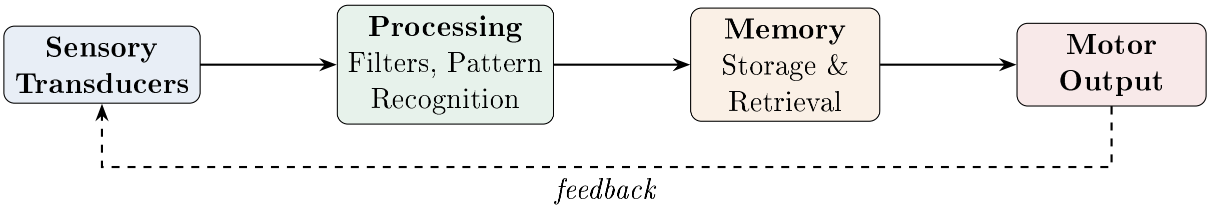
Q8. [Essay] What is the fusiform face area (FFA)? Connect it to distributed representation and to the Capgras syndrome case study.

Answer: The fusiform face area (FFA) is a region in the temporal lobe (fusiform gyrus) that is *specialised* for face processing—it responds strongly and selectively to faces, much more than to objects. Distributed representation link: although the FFA shows high selectivity for faces, full face recognition also engages the occipital lobe (early visual processing), parietal lobe (spatial features), and prefrontal cortex (decision about identity). Capgras link: Capgras patients have an intact FFA (they *recognise* the face) but the pathway to the amygdala (emotional evaluation) is damaged—so recognition is present but the emotional "familiarity" signal is absent. This confirms the two-route model: cognitive recognition vs. emotional evaluation.

Lecture 6: Cognitive Architecture, Memory Taxonomy, and Perception (March 24)

6.1 Architecture of Human Cognition

The brain can be understood as a **hierarchical input–output system**:



Sensory Transducers

Sensory transducers convert external physical signals (photons, pressure waves, chemical molecules, temperature) into the brain's internal electrical signals. Examples:

- **Retina** — converts photons → electrical signals
- **Cochlea (inner ear)** — converts pressure waves → electrical signals
- **Skin receptors** — convert mechanical/thermal stimuli → electrical signals

After transduction, **preliminary analysis** occurs: detecting brightness, edges, foreground vs. background, corners, depth—these are the **basic filters**. This information is then synthesised hierarchically into **patterns**.

The Brain Reconstructs Reality

The brain does not passively record the world. It **internally regenerates its own version of reality** from filtered, transduced signals. What you perceive is a *reconstruction*, not a direct copy of the physical world.

The cognitive architecture also includes:

- **Memory processes** — encoding, storage, and retrieval
- **Thought processes** — reasoning, inference, decision-making
- **Desires, motivations, intentions** — future goals, plans, sub-goal monitoring
- **Working memory** — temporary storage for ongoing tasks
- **Attentional mechanisms** — resource allocation in a parallel system
- **Emotions** — personalisation, sense of self, survival instincts
- **Consciousness/awareness** — self-awareness at the top of the hierarchy

6.2 Serial vs. Parallel Processing

Humans Are Parallel Processors

Unlike digital computers (primarily serial), humans process information **in parallel**. While speaking (serial output), we simultaneously:

- Make hand gestures (motor cortex)
- Monitor visual surroundings (visual cortex)
- Maintain balance (cerebellum)
- Breathe (brainstem)

If we were purely serial, we could not speak and breathe at the same time. **Attention** is the mechanism that allocates resources in this parallel system—focusing on what is relevant while suppressing what is not.

6.3 Emotions in Cognition

Emotions

Emotions are complex psychological and physiological states that influence cognition, memory, and behaviour. Their role includes:

1. **Survival instinct** — primordial fear drives avoidance of danger (self-preservation across all biological systems)
2. **Emotionally-tagged memory** — emotionally significant events are stored more vividly and retrieved more easily
3. **Empathy** — the ability to simulate another person's emotional state ("I can put myself in your shoes")
4. **Sense of self** — emotions personalise experience; they define your likes, dislikes, values, and identity
5. **Approach/avoidance behaviour** — emotions convert neutral stimuli into personally significant ones ("I like this" vs. "I dislike this")

Open Question: Emotion–Cognition Link

The linkage between emotion and cognition is still not fully understood. Active research explores how emotions modulate cognitive processes. A lecture series at IIIT discussed the *neuroscience of emotion and cognition* specifically. Emotion research remains one of the frontiers of cognitive science.

6.4 Consciousness Revisited

- **Consciousness** sits at the top of the cognitive architecture—a “cloud” area still not fully understood
- For decades, consciousness was **banned** from scientific discussion (considered only a spiritual/religious topic)
- In the last 5–10 years, scientific investigation of consciousness has become acceptable
- It remains a major open problem in cognitive science

6.5 Sensation vs. Perception: Top-Down and Bottom-Up

Sensation vs. Perception

- **Sensation** = incoming sensory input being processed (raw data)
- **Perception** = our *interpretation* and *organisation* of that input (making sense of data)

Top-Down vs. Bottom-Up Processing

- **Top-down processing:** Using *prior knowledge*, templates, and prototypes stored in memory to interpret ambiguous stimuli. Example: reading partially occluded text—you fill in the missing letters from your knowledge of English.
- **Bottom-up processing:** The stimulus itself, through the sensory system, drives perception *without* prior knowledge. Example: the Kanizsa illusion—your visual system creates an apparent shape from the arrangement of elements, regardless of your language or cultural knowledge.

Visual Illusions as Evidence

Kanizsa triangle: You perceive a white triangle that is *not explicitly drawn*—your visual system fills in contours from the arrangement of pac-man shapes. This is **bottom-up** (a Russian speaker with no knowledge of English would see the same triangle).
Occluded text: The word “IBC” is partially hidden behind stripes. You can read it because your **top-down** knowledge of English fills in the gaps. A Russian speaker would *not* be able to decode it—demonstrating top-down processing.
Cafe wall illusion: Perfectly straight parallel lines appear sloped/zigzag due to the arrangement of black-and-white tiles. This is **bottom-up**—an artefact of how your visual system processes contrast.

6.6 The Brain as a Predictive Machine

Predictive Processing Hypothesis

The dominant current theory of brain function: the brain is a **probabilistic prediction machine**.
Using **Bayesian inference**:

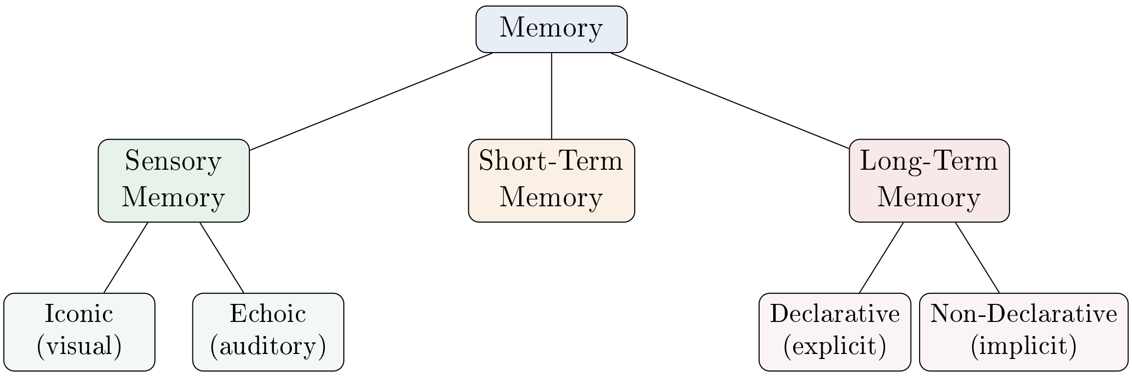
$$P(\text{posterior}) = \frac{P(\text{likelihood}) \times P(\text{prior})}{P(\text{normalisation})}$$

The brain:

1. Maintains a **prior** (what it expects based on past experience)
2. Observes **likelihood** (current sensory evidence)
3. Computes a **posterior** estimate (best guess of current state)
4. Compares prediction with observation; the **difference** = **prediction error**
5. Uses the error to **update the internal model**

This is also called the **predictive coding** hypothesis.
The brain *dislikes ambiguity and uncertainty*—it is always trying to make sense of the world (V.S. Ramachandran).

6.7 Memory Taxonomy (Detailed)



6.7.1 Sensory Memory

- **Duration:** order of seconds (very fleeting)
- **Iconic memory:** visual after-image that persists briefly after stimulus removal
- **Echoic memory:** auditory after-echo (e.g., someone rapidly says a phone number → you can "hear" the echo for a few seconds)
- Erased quickly when new stimuli arrive

6.7.2 Short-Term Memory

- **Duration:** order of minutes to hours
- Example: what you had for breakfast today—you remember it 3 hours later, but likely not tomorrow
- Short-term memories are overwritten by subsequent experiences (lunch erases memory of breakfast)
- Capacity limited (relates to Miller's 7 ± 2)

6.7.3 Long-Term Memory: Declarative (Explicit)

Declarative Memory

Declarative (explicit) memory is memory that can be *consciously recalled and stated.* It has lexical/structural form—you can *explain* what you know.
Two subtypes:

1. **Semantic memory:** knowledge of facts, meanings, and concepts
   - What does "echoic" mean? → Related to "echo" + suffix
   - Who was the first President of India? → Rajendra Prasad
   - Knowledge from textbooks, definitions, general world knowledge
2. **Episodic memory:** vivid recollection of specific personal episodes
   - The day you got your IIIT admission: what you were wearing, who told you, where you were, what time it was
   - Emotionally significant events stored with rich contextual detail

Properties of Declarative Memory

- **Easy to acquire:** repeat something 10 times → memorised (mnemonics help: "OCEAN" for personality types, "All Silver Tea Cups" for trigonometry signs)
- **Easy to lose:** if not rehearsed, declarative memories fade
- Can be *explained* and *communicated* to others

6.7.4 Long-Term Memory: Non-Declarative (Implicit)

Non-Declarative Memory

Non-declarative (implicit) memory cannot be easily verbalised or explained. You *know how* to do something but cannot articulate the precise details.
Two major subtypes:

1. **Procedural (motor) memory:** skills, habits, routines
2. **Priming:** pre-potentiation of the perceptual system

Procedural Memory.

Properties of Procedural Memory

- **Hard to acquire:** riding a bicycle takes a whole month; driving takes ∼2 months
- **Hard to erase:** once learned, you can ride a bicycle years later with no practice ("muscle memory")
- **Non-declarative:** you cannot fully explain *how* you ride (what angle to lean, what force to apply)
- **Transferable:** write your name with your foot in sand even without practice—the motor *program* transfers across effectors
- **Survives neurodegeneration:** Alzheimer's patients who lose declarative memory can still ride a bicycle and do mirror-tracing

Brain areas involved: Motor cortex (central sulcus area), cerebellum, basal ganglia. The field of motor memory acquisition is still not fully understood.

Priming.

Priming

Priming is the **pre-potentiation** of the perceptual/cognitive system so that a future stimulus is recognised more quickly. It operates *below conscious awareness* (subliminally).

- Example: Seeing a bearded man's face → later, when shown an ambiguous image, you immediately see a bearded figure (visual priming)
- Example: Being primed with wedding-related objects → interpreting a neutral story as being about a wedding
- Advertisers heavily exploit priming (subliminal marketing)

- Priming can be visual, auditory, or cross-modal

6.8 The Motor Homunculus

Homunculus

The **homunculus** (Latin: “little man”) is the **topographic map** of the body on the motor and somatosensory cortex. The body parts are represented in proportion to the *amount of cortical area* devoted to their control, **not** in proportion to their physical size.

Key Features of the Homunculus

The representation is **non-linear**:

- **Face** has a disproportionately *large* cortical area (∼200 facial muscles → enables expressive dance, acting, micro-expressions)
- **Hands** have a large cortical area (fine motor control; **opposable thumb**—a key human adaptation separating us from other animals; enables fine grip)
- **Legs** have a relatively *small* cortical area despite being physically large
- Shown on a **coronal section** at the level of the central sulcus
- The same topographic organisation exists for both **motor** and **somatosensory** cortex

Practice & Review

Q1. [Multiple Choice] Non-declarative (implicit) long-term memory includes:

(A) Semantic and episodic memory (B) Sensory and working memory
(C) Procedural (motor skill) memory and priming (D) Short-term memory and iconic memory

Answer: **C.** Non-declarative memory cannot easily be verbalised. It includes procedural memory (“knowing how”—riding a bicycle) and priming (subconscious pre-potentiation of recognition). Semantic and episodic memories are subtypes of *declarative* (explicit) long-term memory.

Q2. [Short Answer] Outline the complete memory taxonomy, including all types and subtypes with approximate durations.

Answer: (1) **Sensory memory** (∼seconds): iconic (visual after-image) and echoic (auditory echo); erased by new input. (2) **Short-term memory** (∼minutes to hours): capacity 7 ± 2 chunks; overwritten by subsequent experiences. (3) **Long-term memory** (days to lifetime): **Declarative (explicit)**—semantic (facts, definitions) and episodic (personal vivid memories with context); **Non-declarative (implicit)**—procedural (motor skills, hard to acquire, hard to erase) and priming (subconscious pre-activation of perception).

Q3. [Essay] Explain the Bayesian brain / predictive processing hypothesis. Write the formula, define each term, and describe the prediction error update cycle.

Answer: The brain is a **probabilistic prediction machine**. Formula:

$$P(\text{posterior}) \propto P(\text{likelihood}) \times P(\text{prior})$$

Prior: the brain's expectation based on past experience. **Likelihood**: the current sensory evidence. **Posterior**: the updated best-guess interpretation after combining prior + evidence. **Prediction error**: the mismatch between the brain's prediction and the actual sensory input. The error signal drives *model updating*: the brain revises its internal model to reduce future prediction errors. This is also called **predictive coding**. Implication: what we perceive is largely our brain's prediction—sensory input merely *corrects* the prediction. The brain "dislikes" ambiguity and uncertainty and constantly seeks the most coherent interpretation (Ramachandran).

Q4. [Multiple Choice] The motor homunculus shows that the body parts with the *largest* cortical representation are:

(A) Legs and trunk (B) Shoulders and back
(C) Face and hands (D) Abdomen and pelvis

Answer: **C.** Face (~200 facial muscles enable rich expression) and hands (fine motor control; opposable thumb is a key human adaptation) have disproportionately large cortical areas. Representation is proportional to *functional importance*, not physical size.

Q5. [Short Answer] Contrast declarative and procedural memory on three dimensions: ease of acquisition, resistance to forgetting, and what happens in Alzheimer's disease.

Answer:

Dimension	**Declarative**	**Procedural**
Acquisition	Easy (repeat ~10 times; mnemonics help)	Hard (weeks–months of practice)
Forgetting	Easy (fades without rehearsal)	Very difficult ("muscle memory" persists)
Alzheimer's	*Lost early* (memory of names, facts)	*Preserved* (can still ride a bicycle)

Q6. [Short Answer] Define priming. Explain the mechanism by which advertisers exploit it.

Answer: Priming is the **pre-potentiation** of the perceptual/cognitive system so that future related stimuli are recognised more rapidly. It operates *below conscious awareness* (subliminally). Advertisers exploit it by repeatedly exposing consumers to brand logos, jingles, or colours in neutral or positive contexts—without the consumer's conscious awareness. Later, encountering the brand activates the pre-potentiated associations (positive feelings, familiarity) and increases likelihood of purchase. Example: a briefly flashed soft-drink logo makes viewers prefer that drink over a competitor's in a subsequent choice.

Q7. [Essay] Explain serial vs. parallel processing in the context of human cognition. How does attention function as a resource allocator in a parallel system?

Answer: **Serial processing** (like digital computers): one process at a time; execution is sequential. **Parallel processing** (human brain): multiple simultaneous processes. Evidence: while speaking, a person simultaneously makes hand gestures (motor cortex), maintains visual monitoring (visual cortex), keeps balance (cerebellum), breathes (brainstem)—all running in parallel. A serial processor could not speak and breathe simultaneously. Problem with parallel processing: *information overload*—all channels compete for the same limited processing resources. **Attention** is the solution: it is the resource-allocation mechanism that selectively amplifies relevant processing channels while suppressing irrelevant ones, enabling efficient behaviour in a massively parallel system.

Q8. [Short Answer] What makes procedural memory clinically unique? Why does it survive Alzheimer's disease?

Answer: Procedural memory is unique because: (1) **Hard to acquire**—requires weeks/months of practice; (2) **Very hard to lose**—remains after years without practice ("once you know how to ride a bicycle, you never forget"); (3) **Non-verbalizable**—you cannot fully explain the precise motor commands used; (4) **Effector-independent**—write your name with your foot or non-dominant hand; the motor *program* transfers. Alzheimer's primarily destroys the **hippocampus** (episodic, declarative memory). Procedural memory is stored in **motor cortex, cerebellum, and basal ganglia**—structures relatively spared in early Alzheimer's. Hence, patients retain motor skills even as they lose factual and autobiographical memories.

Lecture 7: The Motor System — Control, Reflexes, and Skill Learning (March 27)

7.1 Motor Control: Definitions

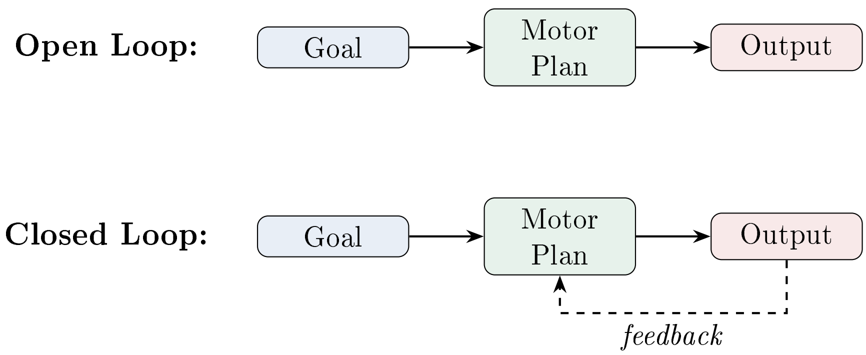
Motor Control

Motor control is the ability to regulate and direct the movements of the body. It is assessed by:

- **Efficiency:** how well a target is hit or a condition is maintained (e.g., bullseye in darts)
- **Robustness/consistency:** how stable performance is across trials (not oscillating wildly)
- **Learning:** repeated practice → measurable improvement in performance scores

The **cerebellum** plays a primary role in motor control and coordination.

7.2 Open-Loop vs. Closed-Loop Control



Open-Loop Control

Open-loop control involves executing a motor action *without* immediate feedback about its outcome. Once initiated, the action runs to completion.
Examples:

- Throwing a football/dart — once released, you cannot adjust mid-flight
- Writing your signature — a practised routine executed without conscious feedback per stroke
- Any well-rehearsed, habitual motor programme

These are typically **memory/habit-driven** actions.

Closed-Loop Control

Closed-loop control involves *continuous feedback* that allows for real-time adjustment of the action.
Examples:

- Driving a car — constantly adjusting speed, direction based on traffic, road conditions
- Threading a needle — perceptual input about needle-eye position guides hand adjustment
- Balancing on an unstable surface

These actions involve a **feedback–adjustment–feedback** cycle.

Many Tasks Are Hybrid

Driving is primarily **closed-loop**, but for an expert on a familiar road, much of it becomes **automatic** (open-loop-like). An unexpected event (construction, rain, oil on road) forces the system back to **closed-loop**. This switch from automatic to deliberate control is an active area of research.

7.3 Handedness

Edinburgh Handedness Inventory

The **Edinburgh Handedness Inventory** is a 100-point questionnaire used by clinicians to determine an individual's **dominant hand**. It asks which hand you use for eating, writing, throwing, cutting, etc.

Handedness and Hemispheric Dominance

- Motor control is **contralateral**: left hemisphere controls right hand and vice versa
- The **left hemisphere is dominant for language** in *most people*—both right-handed *and* left-handed
- Knowing handedness is clinically critical before brain interventions (e.g., surgery for stroke) to identify which hemisphere is more important for the patient
- One evolutionary theory: mothers held infants near the heart (left side) → right hand became free for tool use → right-handedness became dominant
- The world is largely designed for right-handed people (exam desks, scissors, gear shifters, etc.)

7.4 Topographic Organisation and Modularity

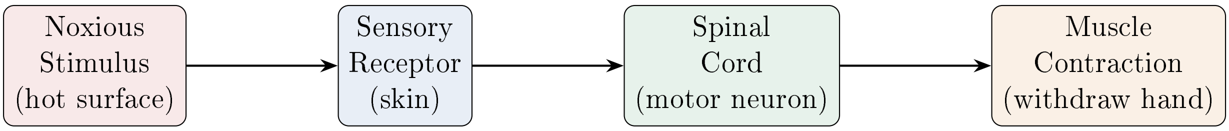
- **Modularity:** different functions (vision, motor, language, auditory) are served by separate brain *modules*
- **Topographic organisation:** within a module, there is a *spatial mapping* of function onto cortical surface (e.g., leg area medial, hand area lateral on motor cortex)
- Not all brain functions are topographically organised, but **sensory and motor functions** are

7.5 Reflexes: The Simplest Motor Circuits

Reflex

A **reflex** is an automatic, involuntary motor response to a sensory stimulus. It does not require conscious processing and is mediated primarily at the **spinal cord** level.

7.5.1 Simple Reflex Arc (Withdrawal Reflex)



Why Reflexes Use the Spinal Cord, Not the Brain

The "long loop" (sensory signal → brain → motor cortex → back down to muscles) would require 4–5 electrochemical hops over a long distance—**too slow** for protective actions. The spinal cord "short circuit" (sensory neuron directly connected to motor neuron at the same spinal level) allows responses within **milliseconds**.

7.5.2 Complex Reflex: Crossed Extensor Reflex

Stepping on a Nail While Walking

If your right foot steps on a nail:

1. **Right leg:** flexor is *excited* (contracts → lifts foot away) + extensor is *inhibited* (relaxes)
2. **Left leg:** extensor is *excited* (stiffens for support) + flexor is *inhibited* (relaxes)

This is the **crossed extensor reflex** (Sherrington, ∼1900). It demonstrates that even "simple" reflexes require **bilateral coordination**—muscles always work in **antagonist pairs** (flexor/extensor).
This built-in circuitry has inspired the design of **humanoid robots** and walking machines.

Knee-Jerk Reflex (Patellar Reflex)

A neurologist taps below the kneecap with a rubber hammer. The leg kicks involuntarily.
Purpose: Tests whether the spinal reflex arc is intact. The signal goes from sensory receptor → spinal cord → motor neuron → quadriceps contraction—all without brain involvement.
If absent → possible spinal cord or peripheral nerve damage.

7.6 Hierarchical Motor Control

Motor control operates at **three hierarchical levels**:

Level	Function	Brain Area
Low level (Execution)	Reflexes, spinal circuits, pattern generators	Spinal cord, brainstem
Mid level (Tactical)	Coordination of movement sequences	Motor cortex, cerebellum
High level (Strategic)	Planning; goal-directed behaviour	Association cortex, prefrontal cortex

At each level there are **feedback loops**. Additionally, there is a distinction between:

- **Unconscious control:** most well-learned actions (walking, breathing, maintaining posture)
- **Conscious control:** new or disrupted situations requiring deliberate attention

7.7 Skill Acquisition: From Conscious to Automatic

Stages of Skill Learning

1. **Early stage (high cognitive load):** Every aspect requires conscious attention. Example: a beginner driver must consciously track mirrors, pedals, gear, steering, traffic—this feels like **sensory overload**. Errors are common (e.g., hitting accelerator instead of brake).

2. **Intermediate stage:** Some actions become semi-automatic; conscious resources begin to free up.

3. **Expert stage (low cognitive load):** Most actions are automatic. The control loop "drops back" to lower hierarchical levels. Cognitive resources are freed—an expert driver can hold a conversation while driving.

Key insight: With expertise, the brain shifts from **conscious, cortical control** to **automatic, subcortical/cerebellar control**. This frees up the prefrontal cortex for other tasks. Identifying the brain areas involved in this transition is an active research area.

7.8 Goal-Directed Behaviour

Goal-Directed Behaviour

Goal-directed behaviour involves *planning* and *executing* a sequence of actions to achieve a specific goal. It requires:

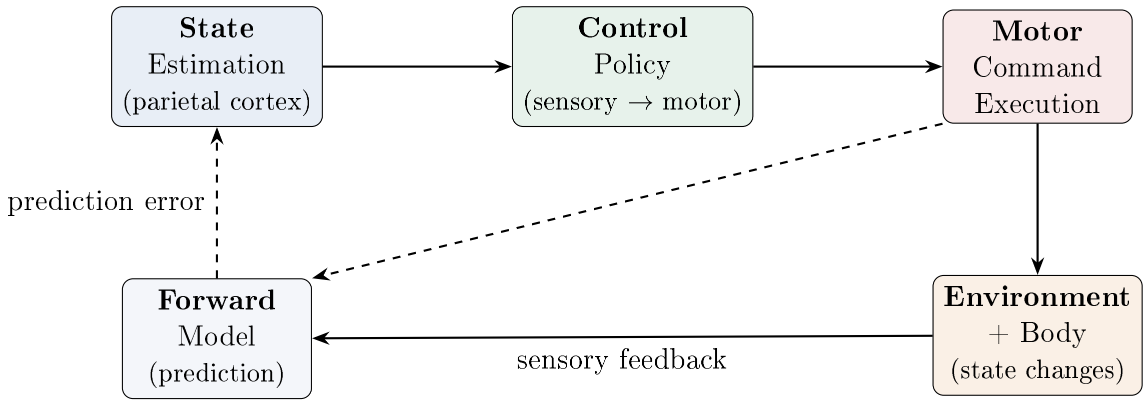
- Decomposing a complex goal into **sub-goals**
- Monitoring progress at each sub-goal
- Adjusting the plan based on feedback

Example: Catching a train from IIIT:

1. Sub-goal 1: Walk/ride to metro station (optimise for time)
2. Sub-goal 2: Ride metro to railway station (optimise: don't stand near door if crowded)
3. Sub-goal 3: Navigate from metro exit to correct platform

Each segment has different optimisation criteria. This requires **conscious, prefrontal** planning.

7.9 The Motor Control Loop: A Control-Systems View



Forward Model

The **forward model** takes a motor command as input and **predicts the expected sensory consequence** of that action *before* sensory feedback arrives. The brain compares this prediction with the actual sensory state; the mismatch is the **prediction error** (or **sensory error**), which drives corrective adjustments.

Inverse Model (Control Policy)

The **inverse model** (or **control policy**) is a mapping from sensory state to appropriate motor action: "Given this sensory state, what motor command should I issue?"

Complete Motor Control Loop

1. **Sense** the current state of the environment (state estimation, parietal cortex)
2. **Decide** on a motor action via the control policy (inverse model)
3. **Execute** the motor command
4. **Predict** the expected sensory consequence (forward model)
5. **Observe** the actual sensory consequence (from environment)
6. **Compute error** = predicted − actual; use error to adjust the policy
7. **Repeat**

Example (driving): You accelerate → forward model predicts you'll be 1m behind the car in front → actual feedback shows you're 5m behind (road is slippery) → error signal adjusts future acceleration commands.

7.10 Cost–Reward Evaluation in Motor Planning

- Humans evaluate **costs and rewards** before executing motor commands

- **Degrees of freedom problem:** there are many ways to perform the same action (e.g., picking up a bottle—overhand, underhand, from the side, with detour around obstacles)
- **Energy cost:** the brain optimises for minimal energy expenditure
- **Smoothness:** dancers optimise for flowing, smooth transitions between movements
- **Accuracy:** surgeons optimise for precise, controlled movements
- These cost–reward considerations inform the control policy and are handled by different brain areas

7.11 Brain–Computer Interfaces (BCIs) for Motor Rehabilitation

Building on the BCI concepts from Lecture 4:

- Conditions like **hemiplegia** (one-sided paralysis) leave the motor *intention* system intact but the motor *output* is cut
- BCI systems decode motor intention from brain activity and drive external devices:
  - **Robotic arms / exoskeletons**
  - **Functional electrical stimulation (FES)** of paralysed muscles
- **Surface-based BCI:** non-invasive EEG electrodes on the scalp
- **Implanted BCI:** electrodes implanted inside the brain (e.g., Neuralink by Elon Musk)
- Current research at IIIT: developing a surface-electrode BCI with the neurosurgery department for stroke patients—reading motor intention from healthy-volunteer data → applying decoders to patients for guided physiotherapy

7.12 Deep Brain Stimulation for Parkinson's Disease (Video Demonstration)

DBS for Parkinson's: Live Demo

A video demonstration was shown in class:

- Patient has a **resting tremor** (hands and limbs shaking involuntarily)
- An electrode is implanted in the **subthalamic nucleus** (part of basal ganglia)
- When the stimulator is switched **ON**: tremor *dramatically reduces* within seconds; patient can move normally
- When switched **OFF**: tremor gradually returns over minutes
- The patient carries a control switch to toggle stimulation

This intervention is used when Parkinson's medication is ineffective for advanced cases. It demonstrates the critical role of the **subthalamic nucleus** in motor control and the power of understanding brain circuitry for clinical intervention.

Practice & Review

Q1. [Multiple Choice] Which type of motor control involves continuous feedback and real-time adjustment of an ongoing action?

(A) Open-loop control (B) Closed-loop control (C) Reflex arc (D) Goal-directed planning

Answer: **B.** Closed-loop control uses a feedback–adjustment–feedback cycle. Open-loop executes a pre-programmed action without mid-course correction (e.g., throwing a dart—once released, the trajectory is fixed).

Q2. [Short Answer] Describe Sherrington's crossed extensor reflex. What does it demonstrate about "simple" reflexes?

Answer: When the right foot steps on a nail: **right leg**—flexor muscles excited (contract, lift foot away) *and* extensor muscles inhibited (relax); **left leg**—extensor muscles excited (stiffen to support body weight) *and* flexors inhibited. This demonstrates: (1) reflexes require **bilateral coordination** across both sides of the body; (2) muscles always work in **antagonist pairs** (flexor/extensor); (3) even "simple" spinal reflexes involve complex circuitry. The circuit has inspired the design of humanoid walking robots.

Q3. [Essay] Describe the complete motor control loop, including the concepts of the forward model, inverse model, and prediction error. Use driving on an icy road as your example.

Answer: The loop has six steps: (1) **State estimation** (parietal cortex): sense the current position and velocity. (2) **Inverse model / control policy**: given the current state, select the appropriate motor command (e.g., how much to accelerate). (3) **Execute** the motor command. (4) **Forward model**: *before* sensory feedback arrives, predict the expected sensory consequence of the action (e.g., "I will be 1m behind the car ahead"). (5) **Observe** actual sensory consequence (e.g., "I am actually 3m behind—the road is more slippery than expected"). (6) **Prediction error** = predicted − actual; this error drives correction of the forward model and the control policy. Driving on ice: the forward model consistently underestimates the slipperiness; prediction errors accumulate and update the model until the driver adapts.

Q4. [Multiple Choice] The Edinburgh Handedness Inventory is a clinical tool used to:

(A) Diagnose motor neuron disease (B) Measure reaction time accuracy
(C) Determine an individual's dominant hand and infer hemispheric dominance
(D) Assess cerebellar function

Answer: **C.** The inventory asks which hand is used for activities (writing, throwing, eating). Clinically critical before brain surgery—identifies which hemisphere controls language (typically left) to avoid damaging it. Note: the left hemisphere dominates for language in both right-handed *and* most left-handed individuals.

Q5. [Short Answer] Name the three hierarchical levels of motor control and the brain areas responsible for each.

Answer: (1) **Low level (Execution)**—spinal cord and brainstem: reflex circuits, central pattern generators (rhythmic locomotion); no conscious involvement. (2) **Mid level (Tactical)**—motor cortex and cerebellum: coordination and execution of movement sequences; error correction. (3) **High level (Strategic)**—association cortex and prefrontal cortex: planning, goal decomposition, sub-goal monitoring; requires conscious attention.

Q6. [Essay] How does skill acquisition change the cognitive load and the neural locus of motor control? Why can an expert driver hold a conversation while driving but a novice cannot?

Answer: Skill acquisition progresses through three stages: (1) **Early (high cognitive load)**: all aspects demand conscious, prefrontal attention—mirrors, pedals, gear, traffic, steering simultaneously. Feels like sensory overload; errors are common. (2) **Intermediate**: some actions become semi-automatic; conscious resources begin to free up. (3) **Expert (low cognitive load)**: most actions are fully automatic, executed by subcortical circuits (motor cortex, cerebellum, basal ganglia). The prefrontal cortex is no longer needed for routine control and is *freed* for other tasks—hence an expert can converse while driving. This transition from cortical → subcortical control is an active research area (identifying the brain areas involved in this "handoff"). Implication: automaticity is a sign of expertise, not inattention.

Q7. [Multiple Choice] The subthalamic nucleus targeted by deep brain stimulation (DBS) for Parkinson's disease is part of the:

(A) Cerebellum (B) Thalamus (C) Basal ganglia (D) Hippocampus

Answer: **C.** The subthalamic nucleus is part of the basal ganglia (the motor output gateway). DBS of this nucleus delivers continuous mild electrical stimulation, normalising aberrant firing patterns and dramatically reducing Parkinson's tremors and rigidity.

Q8. **[Short Answer]** Explain the degrees of freedom problem in motor planning. How does the brain resolve it?

Answer: The same goal (e.g., picking up a water bottle) can be achieved through *many* different movement trajectories (overhand grip, underhand grip, from the side, with detour around an obstacle). The motor system has far more degrees of freedom than necessary—it must select *one* specific trajectory. The brain resolves this via **cost–reward evaluation**: minimising energy expenditure, maximising smoothness (dancers), maximising accuracy (surgeons), or optimising for time. Different optimisation criteria engage different brain areas. Research in motor control studies which cost functions the brain implicitly minimises.

Comprehensive Summary

Supplementary: Handwritten Notes

The file `IBC-26_Mar_2026.pdf` in the `Notes/` folder contains handwritten lecture notes from the March 26 session. These notes provide additional visual diagrams, annotations, and personal insights that complement the typed content in this guide. Refer to them especially for hand-drawn brain diagrams, circuit sketches, and any topic elaborations made in that session that fall between the March 24 and March 27 lectures covered here.

Timeline of Key Ideas

Era	Approach	Central Thesis
1800s	Introspection	Self-observation reveals the mind
1900s–1950s	Behaviourism	Only S–R associations; no "mind"
1950s–60s	Cognitive Revolution	Mind mediates behaviour; internal representations
Emerging	Consciousness studies	Qualia; subjective experience; self-awareness

Brain Organisation at a Glance

Region	Key Functions
Frontal Lobe	Executive function, motor control, speech (Broca's), planning, personality
Parietal Lobe	Somatosensory, proprioception, spatial orientation
Occipital Lobe	Visual processing (object recognition, motion)
Temporal Lobe	Auditory processing, language comprehension (Wernicke's), object naming
Insula	Sense of self, saliency processing
Amygdala	Fear, emotional evaluation
Hippocampus	Long-term memory
Basal Ganglia	Motor gateway
Thalamus	Sensory relay station
Hypothalamus	Homeostasis (hunger, temperature, breathing)
Cerebellum	Motor coordination, balance
Brainstem	Vital functions (respiration, heart rate, sleep–wake)

Key Methods Summary

Method	Measures	Invasive?	Temporal Res.
EEG	Electrical activity (scalp)	No	ms (excellent)
MEG	Magnetic fields (scalp)	No	ms (excellent)
fMRI	Blood oxygenation (BOLD)	No	seconds (poor)
PET	Glucose metabolism	Yes (radiotracer)	minutes (poor)
ECoG	Electrical activity (cortex)	Yes (surgery)	ms (excellent)
DBS	Stimulates subcortical areas	Yes (surgery)	—

Key Numerical Values to Remember

Quantity	Value
Working memory capacity	7 ± 2 chunks
Number of neurons in the brain	$\sim 10^{10}$ (10 billion)
Average connectivity per neuron	$\sim 10^{4}$ (10,000 synapses)
Simple visual reaction time	$\sim$ 100 ms
Object recognition threshold	$\sim$ 100 ms
Batsman response time needed	$\sim$ 15–50 ms
Action potential duration	$\sim$ 1 ms
Action potential amplitude	$\sim$ 30 mV (from -70 to $+30$ mV)
Resting membrane potential	-70 mV
Firing threshold	-55 mV
Synaptic cleft width	$\sim 1\ \mu$m
CNS maturation age	$\sim$ 21 years
Frontal lobe maturation	$\sim$ 18–21 years
ERP signal amplitude	$\sim 2\ \mu$V
Facial muscles	$\sim$ 200
Sensory memory duration	$\sim$ seconds
Short-term memory duration	$\sim$ minutes to hours
Skill learning (bicycle)	$\sim$ 1 month practice
Expert driving acquisition	$\sim$ 2 months practice

Important Case Studies

1. **Phineas Gage** — Iron rod through frontal lobe → personality change, loss of planning/social behaviour
2. **Broca's Aphasia** — Frontal lobe damage → cannot produce speech, can comprehend
3. **Wernicke's Aphasia** — Temporal lobe damage → fluent but nonsensical speech, cannot comprehend
4. **Capgras Syndrome** — Intact recognition but lost emotional evaluation → "imposter" delusion
5. **Alzheimer's Disease** — Temporal lobe neurodegeneration → memory loss, disorientation; procedural memory remains intact
6. **Parkinson's Disease** — Basal ganglia (subthalamic nucleus) dysfunction → resting tremor, rigidity; treated with DBS (dramatic video demonstration)

7. **Cerebellar Ataxia** — Cerebellum damage (e.g., chronic alcoholism) → loss of motor coordination
8. **Hemiplegia** — Motor output cut (e.g., stroke) → motor intention intact but cannot execute; target for BCI rehabilitation
9. **Visual Illusions** — Kanizsa triangle, cafe wall illusion demonstrate bottom-up perceptual construction

Core Concepts Checklist

- ☐ Introspection and its limitations
- ☐ Behaviourism: S–R associations, reflexes, limitations
- ☐ Cognitive Revolution: Tolman (cognitive maps), Miller (7 ± 2), Chomsky (UG)
- ☐ Schemas, scripts, and counterfactual reasoning
- ☐ Consciousness, qualia, unconscious processes
- ☐ CNS vs. PNS; sensory–motor loop
- ☐ Grey matter vs. white matter; cortex vs. spinal cord arrangement
- ☐ CSF: function and location
- ☐ Four lobes: frontal, parietal, occipital, temporal + insula
- ☐ Sulci, gyri, fissures (central sulcus, lateral fissure, etc.)
- ☐ Subcortical structures: amygdala, hippocampus, basal ganglia, thalamus, hypothalamus
- ☐ Cerebellum and brainstem functions
- ☐ Brain development milestones; frontal lobe maturation to age 21
- ☐ PET, MRI, EEG, MEG, fMRI, ECoG, DBS
- ☐ EEG frequency bands: delta, theta, alpha, beta, gamma
- ☐ fMRI BOLD signal; mental imagery experiments
- ☐ Brain–Computer Interfaces (BCIs); surface vs. implanted
- ☐ Neurons: dendrites, soma, axon; action potentials
- ☐ Synapses: electrochemical transmission; synaptic plasticity
- ☐ Hodgkin–Huxley: resting (-70 mV), threshold (-55 mV)
- ☐ Spatial scales (molecules to CNS) and temporal scales (ms to years)

- ☐ Brain vs. computer comparison (parallel vs. sequential, distributed vs. localised, etc.)
- ☐ Grandmother cell vs. distributed coding
- ☐ Nature vs. nurture: both contribute; synaptic refinement through experience
- ☐ Perception as interpretation; visual illusions
- ☐ Sensory transducers: conversion of physical signals to electrical signals
- ☐ Top-down vs. bottom-up processing (Kanizsa triangle, occluded text)
- ☐ Predictive processing / Bayesian brain hypothesis (prior, likelihood, posterior, prediction error)
- ☐ Memory taxonomy: sensory (iconic, echoic), short-term, long-term (declarative vs. non-declarative)
- ☐ Declarative memory: semantic (facts) vs. episodic (personal events)
- ☐ Non-declarative memory: procedural (motor skills, hard to acquire/erase) and priming (subliminal)
- ☐ Motor homunculus: non-linear topographic map (face and hands overrepresented)
- ☐ Emotions: survival instinct, sense of self, personalisation, empathy
- ☐ Open-loop vs. closed-loop motor control
- ☐ Reflexes: withdrawal reflex, crossed extensor reflex (Sherrington), knee-jerk (patellar) reflex
- ☐ Hierarchical motor control: execution (spinal), tactical (motor cortex/cerebellum), strategic (association cortex)
- ☐ Skill acquisition stages: conscious (high cognitive load) → automatic (low cognitive load)
- ☐ Motor control loop: state estimation, control policy, forward model, prediction error
- ☐ Cost–reward evaluation in motor planning; degrees of freedom problem
- ☐ Handedness: Edinburgh Handedness Inventory; left hemisphere dominant for language
- ☐ DBS for Parkinson's: subthalamic nucleus stimulation; dramatic tremor reduction
- ☐ Goal-directed behaviour: sub-goal decomposition, monitoring, adjustment